

Allergy by Proxy to Penicillins Presenting as Recurrent Idiopathic Anaphylaxis Within the Household

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Dataset note: This case section was extracted mechanically from the cited PMC article. No clinical facts were added.

1. Case Report

A 38-year-old man with a history of allergic rhinitis experienced seven episodes of anaphylaxis at home, one of which progressed to anaphylactic shock. He denied any medication use, ingestion of specific foods or Hymenoptera stings prior to these episodes.

The initial evaluation was conducted in an allergist's office. The baseline serum tryptase level was 4.10 g/L. Skin testing for common food allergens yielded negative results, with an appropriate positive histamine control.

The patient was subsequently referred to our unit for further assessment. On detailed re-questioning, he reported that his anaphylactic reactions consistently coincided with periods during which his children were receiving penicillin therapy, and he was able to identify specific triggers for two of these episodes (Table 1).

TABLE 1.

Patient's immediate-type hypersensitivity reactions resulted from indirect exposure to penicillins.

Allergen	Trigger	Onset time	Clinical presentation	Acute serum tryptase	Treatment	Resolution time
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Amoxicillin	Licking his child's spoon containing traces of amoxicillin syrup	5 min				
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	Palmo-plantar itching					
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	Head itching					
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	Facial angioedema					
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	Dyspnea					
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Not measured	Oral corticosteroids	A few hours				
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Amoxicillin	Drinking from his child's glass containing traces of a crushed amoxicillin tablet	10 min				
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	Generalised angioedema					
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Vomiting

Dyspnea

Loss of consciousness

Not measured None A few hours

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Serum-specific IgE testing for beta-lactams (ImmunoCAP, Thermo Fisher Scientific/Phadia, Uppsala, Sweden) was positive only for amoxicillin (0.26 kUA/L). Total serum IgE was not assessed.

Skin testing for beta-lactams was performed in accordance with established guidelines [1]. Skin prick tests produced immediate positive reactions to amoxicillin (0.2 mg/mL; 1:100) and ampicillin (2 mg/mL; 1:10), whereas intradermal tests yielded an immediate positive reaction to penicillin G (10,000 IU/mL; undiluted), confirming sensitisation to penicillins.

The absence of hypersensitivity to alternative beta-lactams—specifically the cephalosporins cefuroxime and ceftriaxone—was demonstrated by negative skin tests (20 mg/mL; undiluted) and negative drug provocation tests at daily therapeutic doses of 1000 and 2000 mg, respectively.